KERALA DIPLOMA CURRICULUM REVISION 2026

Course 4061 | 4 Credits: 4 | Program core | Source-grounded

Yarn Manufacture II

 To impart the knowledge of constructional features and working of Speed frame,

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTR syllabus PDF for Course 4061 is the authoritative source for course information, objectives, course outcomes, CO–PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 4061.pdf from the uploaded official SITTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Textile Technology
Course code	4061
Course title	Yarn Manufacture II
Semester	4 Credits: 4
Course category	Program core
Periods per week	4 (L:3, T:1, P:0) Periods per semester: 60

Item	Official information
Periods per semester	60
Credits	4
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

16 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD

How to use this lesson

First pass

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- To impart the knowledge of constructional features and working of Speed frame,
- Ring frame , doubling machines & advanced spinning systems
- To understand the process involved in the conversion sliver to yarn
- To estimate the draft, production and efficiency of the machines.

Course outcomes

CO	Official outcome	Study level
CO1	18 Applying frame. Describe the process of yarn formation through	See official syllabus
CO2	condensed yarn spinning and other advanced 15 Understanding spinning techniques. Discuss the different methods of yarn doubling	See official syllabus
CO3	10 Understanding techniques and reeling of yarn. Illustrate and solve the problems related to draft,	See official syllabus
CO4	twist, angle of yarn pull, traveler lag and 15 Applying determine production of various machines.	See official syllabus

CO–PO mapping evidence

Row	Extracted official mapping
CO1	CO1 3 1
CO2	CO2 3
CO3	CO3 3
CO4	CO4 3 3

COVERAGE AUDIT

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	Illustrate the working of Speed frame and Ring frame	4	As specified
Module 2	and advanced spinning techniques.	4	As specified
Module 3	of yarn.	4	As specified
Module 4	determine production of speed frame, ring frame, rotor spinning, Ring	4	As specified

MODULE 1

Illustrate the working of Speed frame and Ring frame

This module is presented from the official syllabus scope for Yarn Manufacture II. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Illustrate the working of Speed frame and Ring frame

- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

5 Understanding the functions of the parts involved. Discuss the modern developments in speed

5 Understanding the functions of the parts involved. Discuss the modern developments in speed is an official topic in Module 1 of Yarn Manufacture II. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 5 Understanding the functions of the parts involved. Discuss the modern developments in speed using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 5 understanding the functions of the parts involved. discuss the modern developments in speed should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 5 Understanding the functions of the parts involved. Discuss the modern developments in speed means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-[illegible] 5 Understanding the functions of the parts involved. Discuss the modern developments in speed [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

4 Understanding frame. Describe the working of Ring frame and the

4 Understanding frame. Describe the working of Ring frame and the is an official topic in Module 1 of Yarn Manufacture II. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding frame. Describe the working of Ring frame and the using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding frame. describe the working of ring frame and the should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Understanding frame. Describe the working of Ring frame and the means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 1-□□ 4 Understanding frame. Describe the working of Ring frame and the □□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□.

functions of the parts involved and building 5 Understanding mechanism of Ring frame. Identify and analyse the possible Ring frame

functions of the parts involved and building 5 Understanding mechanism of Ring frame. Identify and analyse the possible Ring frame is an official topic in Module 1 of Yarn Manufacture II. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify functions of the parts involved and building 5 Understanding mechanism of Ring frame. Identify and analyse the possible Ring frame using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, functions of the parts involved and building 5 understanding mechanism of ring frame. identify and analyse the possible ring frame should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What functions of the parts involved and building 5 Understanding mechanism of Ring frame. Identify and analyse the possible Ring frame means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-[illegible] functions of the parts involved and building 5 Understanding mechanism of Ring frame. Identify and analyse the possible Ring frame [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

processing defects and suggest remedies to 4 Applying solve them. Objectives of Speed frame - working of speed frame machine- SKF PK 1500 drafting system - Constructional details of flyer assembly - Function of cots, condenser, false twister, spindle, Principles of winding - Flyer lead and bobbin lead, objectives of builder motion, Objectives of Differential motion, Developments in speed frame - Features of multi motor drive system, Roving stop motion using light barriers, Auto doffing and bobbin transport system. Objectives of Ring frame - working of Ring frame machine - LR pneumatic top arm drafting system- function of bobbin holder, roving traverse guide, fluted and knurled roller, spacer, cradle, nose bar, roller clearer, pneumafil suction system, Traversing thread guide, ABC ring, separator, top roller cots and shore hardness, aprons-Rings Types - Low crown ring & Anti-wedge ring - Comparison of C-traveller and Elliptical traveler - concept of Traveller number and Ring Flange number ,Running in period of ring, Construction details of Modern spindle, Twisting principle, objectives of building mechanism, working of building mechanism of Ring frame for weft wind, Methods for achieving variable spindle speed in spinning - VFD & VPS. Identify and analyze the possible Ring frame processing defects such as excessive end breaks, slubs, thick and thin places, soft yarn, neps, snarls and suggest remedies to solve them. Describe the process of yarn formation through condensed yarn spinning

processing defects and suggest remedies to 4 Applying solve them. Objectives of Speed frame - working of speed frame machine- SKF PK 1500 drafting system - Constructional details of flyer assembly - Function of cots, condenser, false twister, spindle, Principles of winding - Flyer lead and bobbin lead, objectives of builder motion, Objectives of Differential motion, Developments in speed frame - Features of multi motor drive system, Roving stop motion using light barriers, Auto doffing and bobbin transport system. Objectives of Ring frame - working of Ring frame machine - LR pneumatic top arm drafting system- function of bobbin holder, roving traverse guide, fluted and knurled roller, spacer, cradle, nose bar, roller clearer, pneumafil suction system, Traversing thread guide, ABC ring, separator, top roller cots and shore hardness, aprons-Rings Types - Low crown ring & Anti-wedge ring - Comparison of C-traveller and Elliptical traveler - concept of Traveller number and Ring Flange number ,Running in period of ring, Construction details of Modern spindle, Twisting principle, objectives of building mechanism, working of building mechanism of Ring frame for weft wind, Methods for achieving variable spindle speed in spinning - VFD & VPS. Identify and analyze the possible Ring frame processing defects such as excessive end breaks, slubs, thick and thin places, soft yarn, neps, snarls and suggest remedies to solve them. Describe the process of yarn formation through condensed yarn spinning is an

official topic in Module 1 of Yarn Manufacture II. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify processing defects and suggest remedies to 4 Applying solve them. Objectives of Speed frame - working of speed frame machine- SKF PK 1500 drafting system - Constructional details of flyer assembly - Function of cots, condenser, false twister, spindle, Principles of winding - Flyer lead and bobbin lead, objectives of builder motion, Objectives of Differential motion, Developments in speed frame - Features of multi motor drive system, Roving stop motion using light barriers, Auto doffing and bobbin transport system. Objectives of Ring frame - working of Ring frame machine - LR pneumatic top arm drafting system- function of bobbin holder, roving traverse guide, fluted and knurled roller, spacer, cradle, nose bar, roller clearer, pneumafil suction system, Traversing thread guide, ABC ring, separator, top roller cots and shore hardness, aprons-Rings Types - Low crown ring & Anti-wedge ring - Comparison of C-traveller and Elliptical traveler - concept of Traveller number and Ring Flange number ,Running in period of ring, Construction details of Modern spindle, Twisting principle, objectives of building mechanism, working of building mechanism of Ring frame for weft wind, Methods for achieving variable spindle speed in spinning - VFD & VPS. Identify and analyze the possible Ring frame processing defects such as excessive end breaks, slubs, thick and thin places, soft yarn, neps, snarls and suggest remedies to solve them. Describe the process of yarn formation through condensed yarn spinning using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, processing defects and suggest remedies to 4 applying solve them. objectives of speed frame - working of speed frame machine- skf pk 1500 drafting system - constructional details of flyer assembly - function of cots, condenser, false twister, spindle, principles of winding - flyer lead and bobbin lead, objectives of builder motion, objectives of differential motion, developments in speed frame -

features of multi motor drive system, roving stop motion using light barriers, auto doffing and bobbin transport system. objectives of ring frame - working of ring frame machine - lr pneumatic top arm drafting system- function of bobbin holder, roving traverse guide, fluted and knurled roller, spacer, cradle, nose bar, roller clearer, pneumafil suction system, traversing thread guide, abc ring, separator, top roller cots and shore hardness, aprons-rings types - low crown ring & anti-wedge ring - comparison of c-traveller and elliptical traveler - concept of traveller number and ring flange number ,running in period of ring, construction details of modern spindle, twisting principle, objectives of building mechanism, working of building mechanism of ring frame for weft wind, methods for achieving variable spindle speed in spinning - vfd & vps. identify and analyze the possible ring frame processing defects such as excessive end breaks, slubs, thick and thin places, soft yarn, neps, snarls and suggest remedies to solve them. describe the process of yarn formation through condensed yarn spinning should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What processing defects and suggest remedies to 4 Applying solve them. Objectives of Speed frame - working of speed frame machine- SKF PK 1500 drafting system - Constructional details of flyer assembly - Function of cots, condenser, false twister, spindle, Principles of winding - Flyer lead and bobbin lead, objectives of builder motion, Objectives of Differential motion, Developments in speed frame - Features of multi motor drive system, Roving stop motion using light barriers, Auto doffing and bobbin transport system. Objectives of Ring frame - working of Ring frame machine - LR pneumatic top arm drafting system- function of bobbin holder, roving traverse guide, fluted and knurled roller, spacer, cradle, nose bar, roller clearer, pneumafil suction system, Traversing thread guide, ABC ring, separator, top roller cots and shore hardness, aprons-Rings Types - Low crown ring & Anti-wedge ring - Comparison of C-traveller and Elliptical traveler - concept of Traveller number and Ring Flange number ,Running in period of ring, Construction details of Modern spindle, Twisting principle, objectives of building mechanism, working of building mechanism of Ring frame for weft wind, Methods for achieving variable spindle speed in spinning - VFD & VPS. Identify and analyze the possible Ring frame processing defects such as excessive end breaks, slubs, thick and thin places, soft yarn, neps, snarls and suggest remedies to solve them. Describe the process of yarn formation through condensed yarn spinning means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ processing defects and suggest remedies to 4 Applying solve them. Objectives of Speed frame - working of speed frame machine- SKF PK 1500 drafting system - Constructional details of flyer assembly - Function of cots, condenser, false twister, spindle, Principles of winding - Flyer lead and bobbin lead, objectives of builder motion, Objectives of Differential motion, Developments in speed frame - Features of multi motor drive system, Roving stop motion using light barriers, Auto doffing and bobbin transport system. Objectives of Ring frame - working of Ring frame machine - LR pneumatic top arm drafting system- function of bobbin holder, roving traverse guide, fluted and knurled roller, spacer, cradle, nose bar, roller clearer, pneumafil suction system, Traversing thread guide, ABC ring, separator, top roller cots and shore hardness, aprons-Rings Types - Low crown ring & Anti-wedge ring - Comparison of C-traveller and Elliptical traveler - concept of Traveller number and Ring Flange number ,Running in period of ring, Construction details of Modern spindle, Twisting principle, objectives of building mechanism, working of building mechanism of Ring frame for weft wind, Methods for achieving variable spindle speed in spinning - VFD & VPS. Identify and analyze the possible Ring frame processing defects such as excessive end breaks, slubs, thick and thin places, soft yarn, neps, snarls and suggest remedies to solve them. Describe the process of yarn formation through condensed yarn spinning □□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Illustrate the working of Speed frame and Ring frame
              Explain the working of simplex machine and
  M1.01                                                         5
Understanding
              the functions of the parts involved.
              Discuss the modern developments in speed
  M1.02                                                         4
Understanding
              frame.
              Describe the working of Ring frame and the
  M1.03       functions of the parts involved and building      5
Understanding
              mechanism of Ring frame.
              Identify and analyse the possible Ring frame
  M1.04       processing defects and suggest remedies to        4
Applying
              solve them.
Contents:
Objectives of Speed frame - working of speed frame machine- SKF PK 1500 drafting
system - Constructional details of flyer assembly - Function of cots, condenser,
false
twister, spindle, Principles of winding - Flyer lead and bobbin lead, objectives
of builder
motion, Objectives of Differential motion, Developments in speed frame - Features
of multi
motor drive system, Roving stop motion using light barriers, Auto doffing and
bobbin
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Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2

and advanced spinning techniques.

This module is presented from the official syllabus scope for Yarn Manufacture II. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: and advanced spinning techniques.
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

compact spinning and benefits of compact 3 Understanding yarn Discus the principle of open end spinning and

compact spinning and benefits of compact 3 Understanding yarn Discus the principle of open end spinning and is an official topic in Module 2 of Yarn Manufacture II. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify compact spinning and benefits of compact 3 Understanding yarn Discus the principle of open end spinning and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, compact spinning and benefits of compact 3 understanding yarn discus the principle of open end spinning and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What compact spinning and benefits of compact 3 Understanding yarn Discus the principle of open end spinning and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ compact spinning and benefits of compact 3 Understanding yarn Discus the principle of open end spinning and □□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□.

4 Understanding working of Rotor spinning. Illustrate the principle of Friction spinning and

4 Understanding working of Rotor spinning. Illustrate the principle of Friction spinning and is an official topic in Module 2 of Yarn Manufacture II. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding working of Rotor spinning. Illustrate the principle of Friction spinning and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding working of rotor spinning. illustrate the principle of friction spinning and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Understanding working of Rotor spinning. Illustrate the principle of Friction spinning and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ 4 Understanding working of Rotor spinning. Illustrate the principle of Friction spinning and □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

4 Understanding working of DREF II spinning machine. Explain the principle of Air jet spinning and

4 Understanding working of DREF II spinning machine. Explain the principle of Air jet spinning and is an official topic in Module 2 of Yarn Manufacture II. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding working of DREF II spinning machine. Explain the principle of Air jet spinning and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding working of dref ii spinning machine. explain the principle of air jet spinning and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Understanding working of DREF II spinning machine. Explain the principle of Air jet spinning and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ 4 Understanding working of DREF II spinning machine. Explain the principle of Air jet spinning and □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

4 Understanding working of Murata Air jet spinner. Need for Compacting element in condensed yarn spinning - concept of spinning Geometry - Working of Elite compact spinning system - advantages of compact yarn - Working of SIRO spinning-Different advanced spinning systems - principle of open end spinning system- working of Rotor spinning system - Functions of opening roller, transport tube, Rotor, navel, take up tube - Relation between fibre length and rotor diameter, rotor speed - Types of rotor groove - advantages and limitations of rotor spinning over ring spinning - Principle of friction spinning - working of DREF II friction spinning - Principle of air jet spinning - fasciated yarn - working of Murata Air jet spinning - comparison of yarn structure and properties of ring, rotor and air jet yarn. Describe the different mechanism of yarn doubling techniques and reeling

4 Understanding working of Murata Air jet spinner. Need for Compacting element in condensed yarn spinning - concept of spinning Geometry - Working of Elite compact spinning system - advantages of compact yarn - Working of SIRO spinning-Different advanced spinning systems - principle of open end spinning system- working of Rotor spinning system - Functions of opening roller, transport tube, Rotor, navel, take up tube - Relation between fibre length and rotor diameter, rotor speed - Types of rotor groove - advantages and limitations of rotor spinning over ring spinning - Principle of friction spinning - working of DREF II friction spinning - Principle of air jet spinning - fasciated yarn - working of Murata Air jet spinning - comparison of yarn structure and properties of ring, rotor and air jet yarn. Describe the different mechanism of yarn doubling techniques and reeling is an official topic in Module 2 of Yarn Manufacture II. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding working of Murata Air jet spinner. Need for Compacting element in condensed yarn spinning - concept of spinning Geometry - Working of Elite compact spinning system - advantages of compact yarn - Working of SIRO spinning-Different advanced spinning systems - principle of open end spinning system- working of Rotor spinning system - Functions of opening roller, transport tube, Rotor, navel, take up tube - Relation between fibre length and rotor diameter, rotor speed - Types of rotor groove - advantages and limitations of rotor spinning over ring spinning - Principle of friction spinning - working of DREF II friction spinning - Principle of air jet spinning - fasciated yarn - working of Murata Air jet spinning - comparison of yarn structure and properties of ring, rotor and air jet yarn. Describe the different mechanism of yarn doubling techniques and reeling using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding working of murata air jet spinner. need for compacting element in condensed yarn spinning - concept of spinning geometry - working of elite compact spinning system - advantages of compact yarn - working of siro spinning-different advanced spinning systems - principle of open end spinning system- working of rotor spinning system - functions of opening roller, transport tube, rotor, navel, take up tube - relation between fibre length and rotor diameter, rotor speed - types of rotor groove - advantages and limitations of rotor spinning over ring spinning - principle of friction spinning - working of dref ii friction spinning - principle of air jet spinning - fasciated yarn - working of murata air jet spinning - comparison of yarn structure and properties of ring, rotor and air jet yarn. describe the different mechanism of yarn doubling techniques and reeling should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Understanding working of Murata Air jet spinner. Need for Compacting element in condensed yarn spinning - concept of spinning Geometry - Working of Elite compact spinning system - advantages of compact yarn - Working of SIRO spinning-Different advanced spinning systems - principle of open end spinning system- working of Rotor spinning system - Functions of opening roller, transport tube, Rotor, navel, take up tube - Relation between fibre length and rotor diameter, rotor speed - Types of rotor groove - advantages and limitations of rotor spinning over ring spinning - Principle of friction spinning - working of DREF II friction spinning - Principle of air jet spinning - fasciated yarn - working of Murata Air jet spinning - comparison of yarn structure and properties of ring, rotor and air jet yarn. Describe the different mechanism of yarn doubling techniques and reeling means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ 4 Understanding working of Murata Air jet spinner. Need for Compacting element in condensed yarn spinning - concept of spinning Geometry - Working of Elite compact spinning system - advantages of compact yarn - Working of SIRO spinning-Different advanced spinning systems - principle of open end spinning system- working of Rotor spinning system - Functions of opening roller, transport tube, Rotor, navel, take up tube - Relation between fibre length and rotor diameter, rotor speed - Types of rotor groove - advantages and limitations of rotor spinning over ring spinning - Principle of friction spinning - working of DREF II friction spinning - Principle of air jet spinning - fasciated yarn - working of Murata Air jet spinning - comparison of yarn structure and properties of ring, rotor and air jet yarn. Describe the different mechanism of yarn doubling techniques and reeling □□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
and advanced spinning techniques.
             Explain the spinning geometry, mechanism of
  M2.01      compact spinning and benefits of compact             3
Understanding
             yarn
             Discus the principle of open end spinning and
  M2.02                                                           4
Understanding
             working of Rotor spinning.
             Illustrate the principle of Friction spinning and
  M2.03                                                           4
Understanding
             working of DREF II spinning machine.
             Explain the principle of Air jet spinning and
  M2.04                                                           4
Understanding
             working of Murata Air jet spinner.
             Series Test - I                                      1

Contents:
Need for Compacting element in condensed yarn spinning - concept of spinning
Geometry -
Working of Elite compact spinning system - advantages of compact yarn - Working
of
SIRO spinning-Different advanced spinning systems - principle of open end
spinning
system- working of Rotor spinning system - Functions of opening roller, transport
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

# of yarn.

This module is presented from the official syllabus scope for Yarn Manufacture II. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: of yarn.
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

2 Understanding doubling Illustrate the working of Ring doubler and

2 Understanding doubling Illustrate the working of Ring doubler and is an official topic in Module 3 of Yarn Manufacture II. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding doubling Illustrate the working of Ring doubler and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding doubling illustrate the working of ring doubler and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding doubling Illustrate the working of Ring doubler and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 3-□□ 2 Understanding doubling Illustrate the working of Ring doubler and □□□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□.

3 Understanding TFO machine Discuss the types of fancy yarn and working of

3 Understanding TFO machine Discuss the types of fancy yarn and working of is an official topic in Module 3 of Yarn Manufacture II. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding TFO machine Discuss the types of fancy yarn and working of using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding tfo machine discuss the types of fancy yarn and working of should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding TFO machine Discuss the types of fancy yarn and working of means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 3-□□ 3 Understanding TFO machine Discuss the types of fancy yarn and working of □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□.

3 Understanding fancy doubler. Describe the objectives of reeling and the

3 Understanding fancy doubler. Describe the objectives of reeling and the is an official topic in Module 3 of Yarn Manufacture II. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding fancy doubler. Describe the objectives of reeling and the using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding fancy doubler. describe the objectives of reeling and the should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding fancy doubler. Describe the objectives of reeling and the means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 3-[illegible] 3 Understanding fancy doubler. Describe the objectives of reeling and the [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

2 Understanding concept of bundling and baling. Contents Objects of doubling - dry and wet doubling - English and scotch system of wet doubling - Effect of Twist direction - D/S ratio of doubled yarn - Types of doubled yarn - ply, cord, cable, poplin, sewing thread, voile - Types of doubling machine - working of Ring doubler - Objectives of assembly winder - working of TFO - Merits of TFO over Ring doubler - Define fancy yarn - Types of fancy yarn - slub yarn, marl yarn, corkscrew yarn, loop yarn and snarl yarn - Working of Fancy doubler - Objectives of Reeling - working of Plain reeling machine - Concept of bundling and baling - relationship between number of knots in a bundle and count. Illustrate the concept of draft, twist, angle of yarn pull, traveler lag and

2 Understanding concept of bundling and baling. Contents Objects of doubling - dry and wet doubling - English and scotch system of wet doubling - Effect of Twist direction - D/S ratio of doubled yarn - Types of doubled yarn - ply, cord, cable, poplin, sewing thread, voile - Types of doubling machine - working of Ring doubler - Objectives of assembly winder - working of TFO - Merits of TFO over Ring doubler - Define fancy yarn - Types of fancy yarn - slub yarn, marl yarn, corkscrew yarn, loop yarn and snarl yarn - Working of Fancy doubler - Objectives of Reeling - working of Plain reeling machine - Concept of bundling and baling - relationship between number of knots in a bundle and count. Illustrate the concept of draft, twist, angle of yarn pull, traveler lag and is an official topic in Module 3 of Yarn Manufacture II. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding concept of bundling and baling. Contents Objects of doubling - dry and wet doubling - English and scotch system of wet doubling - Effect of Twist direction - D/S ratio of doubled yarn - Types of doubled yarn - ply, cord, cable, poplin, sewing thread, voile - Types of doubling machine - working of Ring doubler - Objectives of assembly winder - working of TFO - Merits of TFO over Ring doubler - Define fancy yarn - Types of fancy yarn - slub yarn, marl yarn, corkscrew yarn, loop yarn and snarl yarn - Working of Fancy doubler - Objectives of Reeling - working of Plain reeling machine - Concept of bundling and baling - relationship between number of knots in a bundle and count. Illustrate the

concept of draft, twist, angle of yarn pull, traveler lag and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding concept of bundling and baling. contents objects of doubling - dry and wet doubling - english and scotch system of wet doubling - effect of twist direction - d/s ratio of doubled yarn - types of doubled yarn - ply, cord, cable, poplin, sewing thread, voile - types of doubling machine - working of ring doubler - objectives of assembly winder - working of tfo - merits of tfo over ring doubler - define fancy yarn - types of fancy yarn - slub yarn, marl yarn, corkscrew yarn, loop yarn and snarl yarn - working of fancy doubler - objectives of reeling - working of plain reeling machine - concept of bundling and baling - relationship between number of knots in a bundle and count. illustrate the concept of draft, twist, angle of yarn pull, traveler lag and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding concept of bundling and baling. Contents Objects of doubling - dry and wet doubling - English and scotch system of wet doubling - Effect of Twist direction - D/S ratio of doubled yarn - Types of doubled yarn - ply, cord, cable, poplin, sewing thread, voile - Types of doubling machine - working of Ring doubler - Objectives of assembly winder - working of TFO - Merits of TFO over Ring doubler - Define fancy yarn - Types of fancy yarn - slub yarn, marl yarn, corkscrew yarn, loop yarn and snarl yarn - Working of Fancy doubler - Objectives of Reeling - working of Plain reeling machine - Concept of bundling and baling - relationship between number of knots in a bundle and count. Illustrate the concept of draft, twist, angle of yarn pull, traveler lag and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ 2 Understanding concept of bundling and baling. Contents Objects of doubling - dry and wet doubling - English and scotch system of wet doubling - Effect of Twist direction - D/S ratio of doubled yarn - Types of doubled yarn - ply, cord, cable, poplin, sewing thread, voile - Types of doubling machine - working of Ring doubler - Objectives of assembly winder - working of TFO - Merits of TFO over Ring doubler - Define fancy yarn - Types of fancy yarn - slub yarn, marl yarn, corkscrew yarn, loop yarn and snarl yarn - Working of Fancy doubler - Objectives of Reeling - working of Plain reeling machine - Concept of bundling and baling - relationship between number of knots in a bundle and count. Illustrate the concept of draft, twist, angle of yarn pull, traveler lag and □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
of yarn.
             Discuss the objectives and the methods of
  M3.01                                                           2
Understanding
             doubling
             Illustrate the working of Ring doubler and
  M3.02                                                           3
Understanding
             TFO machine
             Discuss the types of fancy yarn and working of
  M3.03                                                           3
Understanding
             fancy doubler.

             Describe the objectives of reeling and the
  M3.04                                                           2
Understanding
             concept of bundling and baling.

Contents
Objects of doubling - dry and wet doubling - English and scotch system of wet
doubling -
Effect of Twist direction - D/S ratio of doubled yarn - Types of doubled yarn -
ply, cord,
cable, poplin, sewing thread, voile - Types of doubling machine - working of Ring
doubler -
Objectives of assembly winder - working of TFO - Merits of TFO over Ring doubler
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4

determine production of speed frame, ring frame, rotor spinning, Ring

This module is presented from the official syllabus scope for Yarn Manufacture II. Use the topic cards for learning and the source transcript for exact coverage.

As specified	**CO-linked**
official hours	see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: determine production of speed frame, ring frame, rotor spinning, Ring
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

4 Applying and machine efficiency of speed frame Determine the TPI, actual draft, angle of yarn

4 Applying and machine efficiency of speed frame Determine the TPI, actual draft, angle of yarn is an official topic in Module 4 of Yarn Manufacture II. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Applying and machine efficiency of speed frame Determine the TPI, actual draft, angle of yarn using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 applying and machine efficiency of speed frame determine the tpi, actual draft, angle of yarn should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Applying and machine efficiency of speed frame Determine the TPI, actual draft, angle of yarn means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ 4 Applying and machine efficiency of speed frame Determine the TPI, actual draft, angle of yarn □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

pull, traveller speed, traveller lag, and twist 4 Applying contraction in Ring frame. Determine the production and machine

pull, traveller speed, traveller lag, and twist 4 Applying contraction in Ring frame. Determine the production and machine is an official topic in Module 4 of Yarn Manufacture II. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify pull, traveller speed, traveller lag, and twist 4 Applying contraction in Ring frame. Determine the production and machine using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, pull, traveller speed, traveller lag, and twist 4 applying contraction in ring frame. determine the production and machine should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What pull, traveller speed, traveller lag, and twist 4 Applying contraction in Ring frame. Determine the production and machine means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ pull, traveller speed, traveller lag, and twist 4 Applying contraction in Ring frame. Determine the production and machine □□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□.

efficiency of Ring frame and Rotor spinning 4 Applying machine. Determine the resultant count , production and

efficiency of Ring frame and Rotor spinning 4 Applying machine. Determine the resultant count , production and is an official topic in Module 4 of Yarn Manufacture II. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify efficiency of Ring frame and Rotor spinning 4 Applying machine. Determine the resultant count , production and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, efficiency of ring frame and rotor spinning 4 applying machine. determine the resultant count , production and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What efficiency of Ring frame and Rotor spinning 4 Applying machine. Determine the resultant count , production and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ efficiency of Ring frame and Rotor spinning 4 Applying machine. Determine the resultant count , production and □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

efficiency of Ring doubler and TFO 3 Applying machine Roving hank calculation - TPI and actual draft calculation of speed frame & ring frame - concept of angle of yarn pull - traveler speed - traveller lag - twist contraction and its calculation - production and efficiency of speed frame & ring frame - production & efficiency of Rotor spinning machine - Resultant count - production and efficiency of Ring doubler & TFO machine. Text / Reference: T/R Book Title/Author T1 W. Klein, Rieter Manual of Spinning Volume 4-6 , Rieter, 2010 R. V. M. Gowda, New Spinning Systems, NCUTE Publication, New Delhi, R2 2005 R3 Oxtoby E., "Spun Yarn Technology", Butterworths, London, 2002. R4 Dr.A.R.Khare- Elements of Ring Frame and Doubling Chattopadhyay R, "Advances in Technology of Yarn Production", NCUTE, IIT R5 Delhi, 2002. Online resources: Sl.No Website Link 1 https://nptel.ac.in 2 www.reiter.com 3 www.laksmimach.com 4 www.suessen.com 5 www.fibre2fashion.com

efficiency of Ring doubler and TFO 3 Applying machine Roving hank calculation - TPI and actual draft calculation of speed frame & ring frame - concept of angle of yarn pull - traveler speed - traveller lag - twist contraction and its calculation - production and efficiency of speed frame & ring frame - production & efficiency of Rotor spinning machine - Resultant count - production and efficiency of Ring doubler & TFO machine. Text / Reference: T/R Book Title/Author T1 W. Klein, Rieter Manual of Spinning Volume 4-6 , Rieter, 2010 R. V. M. Gowda, New Spinning Systems, NCUTE Publication, New Delhi, R2 2005 R3 Oxtoby E., "Spun Yarn Technology", Butterworths, London, 2002. R4 Dr.A.R.Khare- Elements of Ring Frame and Doubling Chattopadhyay R, "Advances in Technology of Yarn Production", NCUTE, IIT R5 Delhi, 2002. Online resources: Sl.No Website Link 1 https://nptel.ac.in 2 www.reiter.com 3 www.laksmimach.com 4 www.suessen.com 5 www.fibre2fashion.com is an official topic in Module 4 of Yarn Manufacture II. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify efficiency of Ring doubler and TFO 3 Applying machine Roving hank calculation - TPI and actual draft calculation of speed frame & ring frame - concept of angle of yarn pull - traveler speed - traveller lag - twist contraction and its calculation - production and efficiency of speed frame & ring frame - production

& efficiency of Rotor spinning machine - Resultant count - production and efficiency of Ring doubler & TFO machine. Text / Reference: T/R Book Title/Author T1 W. Klein, Rieter Manual of Spinning Volume 4-6 , Rieter, 2010 R. V. M. Gowda, New Spinning Systems, NCUTE Publication, New Delhi, R2 2005 R3 Oxtoby E., “Spun Yarn Technology”, Butterworths, London, 2002. R4 Dr.A.R.Khare- Elements of Ring Frame and Doubling Chattopadhyay R, “Advances in Technology of Yarn Production”, NCUTE, IIT R5 Delhi, 2002. Online resources: Sl.No Website Link 1 https://nptel.ac.in 2 www.reiter.com 3 www.laksmimach.com 4 www.suessen.com 5 www.fibre2fashion.com using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, efficiency of ring doubler and tfo 3 applying machine roving hank calculation - tpi and actual draft calculation of speed frame & ring frame - concept of angle of yarn pull - traveler speed - traveller lag - twist contraction and its calculation - production and efficiency of speed frame & ring frame - production & efficiency of rotor spinning machine - resultant count - production and efficiency of ring doubler & tfo machine. text / reference: t/r book title/author t1 w. klein, rieter manual of spinning volume 4-6 , rieter, 2010 r. v. m. gowda, new spinning systems, ncute publication, new delhi, r2 2005 r3 oxtoby e., “spun yarn technology”, butterworths, london, 2002. r4 dr.a.r.khare- elements of ring frame and doubling chattopadhyay r, “advances in technology of yarn production”, ncute, iit r5 delhi, 2002. online resources: sl.no website link 1 https://nptel.ac.in 2 www.reiter.com 3 www.laksmimach.com 4 www.suessen.com 5 www.fibre2fashion.com should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What efficiency of Ring doubler and TFO 3 Applying machine Roving hank calculation - TPI and actual draft calculation of speed frame & ring frame - concept of angle of yarn pull - traveler speed - traveller lag - twist contraction and its calculation - production and efficiency of speed frame & ring frame - production & efficiency of Rotor spinning machine - Resultant count - production and efficiency of Ring doubler & TFO machine. Text / Reference: T/R Book Title/Author T1 W. Klein, Rieter Manual of Spinning Volume 4-6 , Rieter, 2010 R. V. M. Gowda, New Spinning Systems, NCUTE Publication, New Delhi, R2 2005 R3 Oxtoby E., “Spun Yarn Technology”, Butterworths, London, 2002. R4 Dr.A.R.Khare- Elements of Ring Frame and Doubling Chattopadhyay R, “Advances in Technology of Yarn Production”, NCUTE, IIT R5 Delhi, 2002. Online resources: Sl.No Website Link 1 https://nptel.ac.in 2 www.reiter.com 3 www.laksmimach.com 4 www.suessen.com 5 www.fibre2fashion.com means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ efficiency of Ring doubler and TFO 3 Applying machine Roving hank calculation - TPI and actual draft calculation of speed frame & ring frame - concept of angle of yarn pull - traveler speed - traveller lag - twist contraction and its calculation - production and efficiency of speed frame & ring frame - production & efficiency of Rotor spinning machine - Resultant count - production and efficiency of Ring doubler & TFO machine. Text / Reference: T/R Book Title/Author T1 W. Klein, Rieter Manual of Spinning Volume 4-6 , Rieter, 2010 R. V. M. Gowda, New Spinning Systems, NCUTE Publication, New Delhi, R2 2005 R3 Oxtoby E., “Spun Yarn Technology”, Butterworths, London, 2002. R4 Dr.A.R.Khare- Elements of Ring Frame and Doubling Chattopadhyay R, “Advances in Technology of Yarn Production”, NCUTE, IIT R5 Delhi, 2002. Online resources: Sl.No Website Link 1 https://nptel.ac.in 2 www.reiter.com 3 www.laksmimach.com 4 www.suessen.com 5 www.fibre2fashion.com □□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
determine production of speed frame, ring frame, rotor spinning, Ring
            doubler and TFO machine.
            Determine the TPI, actual draft, production
  M4.01                                                        4
Applying
            and machine efficiency of speed frame
            Determine the TPI, actual draft, angle of yarn
  M4.02     pull, traveller speed, traveller lag, and twist      4
Applying
            contraction in Ring frame.
            Determine the production and machine
  M4.03     efficiency of Ring frame and Rotor spinning          4
Applying
            machine.
            Determine the resultant count , production and
  M4.04     efficiency of Ring doubler and TFO                   3
Applying
            machine
            Series Test - II                                     1
Contents:
Roving hank calculation - TPI and actual draft calculation of speed frame & ring
frame -
concept of angle of yarn pull - traveler speed - traveller lag - twist
contraction and its
calculation - production and efficiency of speed frame & ring frame - production
&
efficiency of Rotor spinning machine - Resultant count - production and
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE

Concept and formula bank

Answer structure

```
Definition → Principle → Components/steps → Application → Conclusion
```

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

```
Aim → Apparatus → Procedure → Observation → Result → Safety
```

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION

Diagram and source-transcript library

Module 1 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING

Activities from the syllabus

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

```
The official assessment structure is reproduced from the supplied PDF.
```

Practice-paper notice: The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

Define or explain 5 Understanding the functions of the parts involved. Discuss the modern developments in speed. (3 marks)

Answer: Begin with the standard definition or identity of *5 Understanding the functions of the parts involved. Discuss the modern developments in speed*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□□.

Define or explain 4 Understanding frame. Describe the working of Ring frame and the. (3 marks)

Answer: Begin with the standard definition or identity of *4 Understanding frame. Describe the working of Ring frame and the*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□□.

Define or explain functions of the parts involved and building 5 Understanding mechanism of Ring frame. Identify and analyse the possible Ring frame. (3 marks)

Answer: Begin with the standard definition or identity of *functions of the parts involved and building 5 Understanding mechanism of Ring frame. Identify and analyse the possible Ring frame*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□□.

Define or explain processing defects and suggest remedies to 4 Applying solve them. Objectives of Speed frame - working of speed frame machine- SKF PK 1500 drafting system - Constructional details of flyer assembly - Function of cots, condenser, false twister, spindle, Principles of winding - Flyer lead and bobbin lead, objectives of builder motion, Objectives of Differential motion, Developments in speed frame - Features of multi motor drive system, Roving stop motion using light barriers, Auto doffing and bobbin transport system. Objectives of Ring frame - working of Ring frame machine - LR pneumatic top arm drafting system- function of bobbin holder, roving traverse guide, fluted and knurled roller, spacer, cradle, nose bar, roller clearer, pneumafil suction

system, Traversing thread guide, ABC ring, separator, top roller cots and shore hardness, aprons-Rings Types - Low crown ring & Anti-wedge ring - Comparison of C-traveller and Elliptical traveler - concept of Traveller number and Ring Flange number ,Running in period of ring, Construction details of Modern spindle, Twisting principle, objectives of building mechanism, working of building mechanism of Ring frame for weft wind, Methods for achieving variable spindle speed in spinning - VFD & VPS. Identify and analyze the possible Ring frame processing defects such as excessive end breaks, slubs, thick and thin places, soft yarn, neps, snarls and suggest remedies to solve them. Describe the process of yarn formation through condensed yarn spinning. (3 marks)

Answer: Begin with the standard definition or identity of *processing defects and suggest remedies to 4 Applying solve them. Objectives of Speed frame - working of speed frame machine- SKF PK 1500 drafting system - Constructional details of flyer assembly - Function of cots, condenser, false twister, spindle, Principles of winding - Flyer lead and bobbin lead, objectives of builder motion, Objectives of Differential motion, Developments in speed frame - Features of multi motor drive system, Roving stop motion using light barriers, Auto doffing and bobbin transport system. Objectives of Ring frame - working of Ring frame machine - LR pneumatic top arm drafting system-function of bobbin holder, roving traverse guide, fluted and knurled roller, spacer, cradle, nose bar, roller clearer, pneumafil suction system, Traversing thread guide, ABC ring, separator, top roller cots and shore hardness, aprons-Rings Types - Low crown ring & Anti-wedge ring - Comparison of C-traveller and Elliptical traveler - concept of Traveller number and Ring Flange number ,Running in period of ring, Construction details of Modern spindle, Twisting principle, objectives of building mechanism, working of building mechanism of Ring frame for weft wind, Methods for achieving variable spindle speed in spinning - VFD & VPS. Identify and analyze the possible Ring frame processing defects such as excessive end breaks, slubs, thick and thin places, soft yarn, neps, snarls and suggest remedies to solve them. Describe the process of yarn formation through condensed yarn spinning*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Module 2

Define or explain compact spinning and benefits of compact 3 Understanding yarn Discus the principle of open end spinning and. (3 marks)

Answer: Begin with the standard definition or identity of *compact spinning and benefits of compact 3 Understanding yarn Discus the principle of open end spinning and*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□ □□□□□□□□□□ □□□□□□□.

Define or explain 4 Understanding working of Rotor spinning. Illustrate the principle of Friction spinning and. (3 marks)

Answer: Begin with the standard definition or identity of *4 Understanding working of Rotor spinning. Illustrate the principle of Friction spinning and*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□ □□□□□□□□□□ □□□□□□□.

Define or explain 4 Understanding working of DREF II spinning machine. Explain the principle of Air jet spinning and. (3 marks)

Answer: Begin with the standard definition or identity of *4 Understanding working of DREF II spinning machine. Explain the principle of Air jet spinning and*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□ □□□□□□□□□□ □□□□□□□.

Define or explain 4 Understanding working of Murata Air jet spinner. Need for Compacting element in condensed yarn spinning - concept of spinning Geometry - Working of Elite compact spinning system - advantages of compact yarn - Working of SIRO spinning-Different advanced spinning systems - principle of open end spinning system- working of Rotor spinning system -

Functions of opening roller, transport tube, Rotor, navel, take up tube - Relation between fibre length and rotor diameter, rotor speed - Types of rotor groove - advantages and limitations of rotor spinning over ring spinning - Principle of friction spinning - working of DREF II friction spinning - Principle of air jet spinning - fasciated yarn - working of Murata Air jet spinning - comparison of yarn structure and properties of ring, rotor and air jet yarn. Describe the different mechanism of yarn doubling techniques and reeling. (3 marks)

Answer: Begin with the standard definition or identity of *4 Understanding working of Murata Air jet spinner. Need for Compacting element in condensed yarn spinning - concept of spinning Geometry - Working of Elite compact spinning system - advantages of compact yarn - Working of SIRO spinning-Different advanced spinning systems - principle of open end spinning system- working of Rotor spinning system - Functions of opening roller, transport tube, Rotor, navel, take up tube - Relation between fibre length and rotor diameter, rotor speed - Types of rotor groove - advantages and limitations of rotor spinning over ring spinning - Principle of friction spinning - working of DREF II friction spinning - Principle of air jet spinning - fasciated yarn - working of Murata Air jet spinning - comparison of yarn structure and properties of ring, rotor and air jet yarn. Describe the different mechanism of yarn doubling techniques and reeling*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Module 3

Define or explain 2 Understanding doubling Illustrate the working of Ring doubler and. (3 marks)

Answer: Begin with the standard definition or identity of *2 Understanding doubling Illustrate the working of Ring doubler and*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 3 Understanding TFO machine Discuss the types of fancy yarn and working of. (3 marks)

Answer: Begin with the standard definition or identity of *3 Understanding TFO machine Discuss the types of fancy yarn and working of*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 3 Understanding fancy doubler. Describe the objectives of reeling and the. (3 marks)

Answer: Begin with the standard definition or identity of *3 Understanding fancy doubler. Describe the objectives of reeling and the*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 2 Understanding concept of bundling and baling. Contents Objects of doubling - dry and wet doubling - English and scotch system of wet doubling - Effect of Twist direction - D/S ratio of doubled yarn - Types of doubled yarn - ply, cord, cable, poplin, sewing thread, voile - Types of doubling machine - working of Ring doubler - Objectives of assembly winder - working of TFO - Merits of TFO over Ring doubler - Define fancy yarn - Types of fancy yarn - slub yarn, marl yarn, corkscrew yarn, loop yarn and snarl yarn - Working of Fancy doubler - Objectives of Reeling - working of Plain reeling machine - Concept of bundling and baling - relationship between number of knots in a bundle and count. Illustrate the concept of draft, twist, angle of yarn pull, traveler lag and. (3 marks)

Answer: Begin with the standard definition or identity of *2 Understanding concept of bundling and baling. Contents Objects of doubling - dry and wet doubling - English and scotch system of wet doubling - Effect of Twist direction - D/S ratio of doubled yarn - Types of doubled yarn - ply, cord, cable, poplin, sewing thread, voile - Types of doubling* *machine - working of Ring doubler - Objectives of assembly winder - working of TFO - Merits of TFO over Ring doubler - Define fancy yarn - Types of fancy yarn - slub yarn, marl yarn, corkscrew yarn, loop yarn and snarl yarn - Working of Fancy doubler - Objectives of Reeling - working of Plain reeling machine - Concept of bundling and baling - relationship between number of knots in a bundle and count. Illustrate the concept of draft, twist, angle of yarn pull, traveler lag and*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□ □□□□□□□□□□ □□□□□□□.

Module 4

Define or explain 4 Applying and machine efficiency of speed frame Determine the TPI, actual draft, angle of yarn. (3 marks)

Answer: Begin with the standard definition or identity of *4 Applying and machine efficiency of speed frame Determine the TPI, actual draft, angle of yarn*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□ □□□□□□□□□□ □□□□□□□.

Define or explain pull, traveller speed, traveller lag, and twist 4 Applying contraction in Ring frame. Determine the production and machine. (3 marks)

Answer: Begin with the standard definition or identity of *pull, traveller speed, traveller lag, and twist 4 Applying contraction in Ring frame. Determine the production and machine*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□ □□□□□□□□□□ □□□□□□□.

Define or explain efficiency of Ring frame and Rotor spinning 4 Applying machine. Determine the resultant count , production and. (3 marks)

Answer: Begin with the standard definition or identity of *efficiency of Ring frame and Rotor spinning 4 Applying machine. Determine the resultant count , production and*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□ □□□□□□□□□□ □□□□□□□.

Define or explain efficiency of Ring doubler and TFO 3 Applying machine Roving hank calculation - TPI and actual draft calculation of speed frame & ring frame - concept of angle of yarn pull - traveler speed - traveller lag - twist contraction and its calculation - production and efficiency of speed frame & ring frame - production & efficiency of Rotor spinning machine - Resultant count - production and efficiency of Ring doubler & TFO machine. Text / Reference: T/R Book Title/Author T1 W. Klein, Rieter Manual of Spinning Volume 4-6 , Rieter, 2010 R. V. M. Gowda, New Spinning Systems, NCUTE Publication, New Delhi, R2 2005 R3 Oxtoby E., "Spun Yarn Technology", Butterworths, London, 2002. R4 Dr.A.R.Khare- Elements of Ring Frame and Doubling Chattopadhyay R, "Advances in Technology of Yarn Production", NCUTE, IIT R5 Delhi, 2002. Online resources: Sl.No Website Link 1 https://nptel.ac.in 2 www.reiter.com 3 www.laksmimach.com 4 www.suessen.com 5 www.fibre2fashion.com. (3 marks)

Answer: Begin with the standard definition or identity of *efficiency of Ring doubler and TFO 3 Applying machine Roving hank calculation - TPI and actual draft calculation of speed frame & ring frame - concept of angle of yarn pull - traveler speed - traveller lag - twist contraction and its calculation - production and efficiency of speed frame & ring frame - production & efficiency of Rotor spinning machine - Resultant count - production and efficiency of Ring doubler & TFO machine. Text / Reference: T/R Book Title/Author T1 W. Klein, Rieter Manual of Spinning Volume 4-6 , Rieter, 2010 R. V. M. Gowda, New Spinning Systems, NCUTE Publication, New Delhi, R2 2005 R3 Oxtoby E., "Spun Yarn Technology", Butterworths, London, 2002. R4 Dr.A.R.Khare- Elements of Ring Frame and Doubling Chattopadhyay R, "Advances in Technology of Yarn Production", NCUTE, IIT R5 Delhi, 2002. Online resources: Sl.No Website Link 1 https://nptel.ac.in 2 www.reiter.com 3 www.laksmimach.com 4 www.suessen.com 5 www.fibre2fashion.com*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **5 Understanding the functions of the parts involved. Discuss the modern developments in speed**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **compact spinning and benefits of compact 3 Understanding yarn Discus the principle of open end spinning and**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **2 Understanding doubling Illustrate the working of Ring doubler and**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 4

1-mark definition: State the meaning of **4 Applying and machine efficiency of speed frame Determine the TPI, actual draft, angle of yarn**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

- Sl.No Website Link https://nptel.ac.in www.reiter.com www.laksmimach.com www.suessen.com www.fibre2fashion.com

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